ACM 104 HW #4

Om Sanan

1. Least Squares Solution

$$A^T = \begin{bmatrix} 1 & 0 & 1 & -3 \\ 2 & -2 & 5 & 1 \\ -1 & 3 & -1 & 1 \end{bmatrix}, \quad A^TA = \begin{bmatrix} 11 & 4 & -5 \\ 4 & 34 & -12 \\ -5 & -12 & 12 \end{bmatrix}, \quad A^TB = \begin{bmatrix} -18 \\ 28 \\ 17 \end{bmatrix}$$

$$A^TA\hat{x} = A^Tb : \begin{bmatrix} 11 & 4 & -5 \\ 4 & 34 & -12 \\ -5 & -12 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -18 \\ 28 \\ 17 \end{bmatrix}$$

$11x_1 + 4x_2 - 5x_3 = -18$ ①

$4x_1 + 34x_2 - 12x_3 = 28$ ② $\rightarrow 2x_1 + 17x_2 - 6x_3 = 14$ 2.5

$-5x_1 - 12x_2 + 12x_3 = 17$ ③

② + ③:

$-x_1 + 22x_2 = 45$ ④

$5 \cdot 2.5 - 6 \cdot ①$

$-56x_1 + 61x_2 = 178$ ⑤

$56 \cdot ④ + ⑤$

$x_2 = 2, x_1 = -1$

Plug in ①: $x_3 = 3$

$$\Rightarrow \boxed{\hat{x} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}}$$

$$A\hat{x} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 1 & 5 & -1 \\ -3 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 6 \\ 8 \end{bmatrix}$$

$$r = b - A\hat{x} = \begin{bmatrix} 0 \\ 5 \\ 6 \\ 8 \end{bmatrix} - \begin{bmatrix} 0 \\ 5 \\ 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\boxed{\text{Error} = \|r\| = 0}$$

2. Interpolation for Integration

a) $p_1(x) = f(a)\dfrac{x-b}{a-b} + f(b)\dfrac{x-a}{b-a}$

$$\int_a^b p_1(x)\,dx = \left(f(a)+f(b)\right)\int_a^b \frac{x-b}{a-b} + \frac{x-a}{b-a}\,dx$$

$$= \frac{f(a)}{a-b}\int_a^b x-b\,dx + \frac{f(b)}{b-a}\int_a^b x-a\,dx$$

$$= \left(\frac{f(a)}{a-b}\right)\left(-\frac{(a-b)^2}{2}\right) + \frac{f(b)}{b-a}\left(\frac{(b-a)^2}{2}\right)$$

$$\boxed{\int_a^b p_1(x)\,dx = \left(f(a)+f(b)\right)\left(\frac{b-a}{2}\right)}$$

b) $x_0 = a + \frac{1}{3}(b-a) = \dfrac{2a+b}{3}$, $x_1 = a + \frac{2}{3}(b-a) = \dfrac{a+2b}{3}$

$$p_1(x) = f(x_0)\frac{x-x_1}{x_0-x_1} + f(x_1)\frac{x-x_0}{x_1-x_0}$$

$$\int p_1(x)\,dx = \int \frac{3}{b-a}\left(f(x_0)(x_1-x) + f(x_1)(x-x_0)\right)dx$$

$$\int_a^b (x_1-x)\,dx = x_1(b-a) - \frac{b^2-a^2}{2} = (b-a)\left(x_1 - \frac{a+b}{2}\right)$$

$$= (b-a)\left(\frac{a+2b}{3} - \frac{a+b}{2}\right) = \frac{(b-a)^2}{6}$$

$$\int_a^b (x-x_0)\,dx = \frac{(b-a)^2}{6} \quad \text{similarly}$$

$$\therefore \int_a^b p_1(x)\,dx = \left(\frac{3}{b-a}\right)\left(\frac{(b-a)^2}{6}\right)\left(f(x_0)+f(x_1)\right) = \boxed{\frac{b-a}{2}\left(f(x_0)+f(x_1)\right)}$$

c) $\int_0^1 e^x\,dx$

Trapezoid: $\frac{1}{2}(e^0+e^1) = \frac{1+e}{2} \approx 1.86$ $\quad E \approx 0.14$

Open: $x_0 = 1/3$, $x_1 = 2/3$

$\frac{1}{2}(e^{1/3}+e^{2/3}) \approx 1.67 \rightarrow E \approx 0.05$

$\int_0^\pi \sin x\,dx$

Trapezoid: $\frac{\pi}{2}(\sin(0)+\sin(\pi)) = 0$ $\quad E = 2$

Open: $\frac{\pi}{2}\left(\sin\left(\frac{\pi}{3}\right)+\sin\left(\frac{2\pi}{3}\right)\right) = \frac{\pi\sqrt{3}}{2}$

$\approx 2.72 \rightarrow E \approx 0.72$

$\boxed{\text{Open rule better.}}$

3. Application of Least Squares to Data Fitting

a) Goal: Solve $(A^TA)B^* = A^Ty$

Model per data pt:

$$y^{(i)} \approx \beta_0 + \beta_1 x_1^{(i)} + \beta_2 x_2^{(i)} + \beta_3 x_1^{(i)} x_2^{(i)}$$

This can be written as linear system:

$$\begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} & x_1^{(1)} x_2^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} & x_1^{(2)} x_2^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_1^{(m)} & x_2^{(m)} & x_1^{(m)} x_2^{(m)} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \approx \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix}$$

$r = y - AB$; now we will find best B^* in $\|r\|^2 = \|y - A\beta\|^2$

$$\boxed{A^TA = \begin{bmatrix} 406 & \sum x_1^{(i)} & \sum x_2^{(i)} & \sum x_1^{(i)} x_2^{(i)} \\ \sum x_1^{(i)} & \sum (x_1^{(i)})^2 & \sum x_1^{(i)} x_2^{(i)} & \sum (x_1^{(i)})^2 x_2^{(i)} \\ \sum x_2^{(i)} & \sum x_1^{(i)} x_2^{(i)} & \sum (x_2^{(i)})^2 & \sum x_1^{(i)} (x_2^{(i)})^2 \\ \sum x_1^{(i)} x_2^{(i)} & \sum (x_1^{(i)})^2 x_2^{(i)} & \sum x_1^{(i)} (x_2^{(i)})^2 & \sum (x_1^{(i)} x_2^{(i)})^2 \end{bmatrix} \begin{bmatrix} \beta_0^* \\ \beta_1^* \\ \beta_2^* \\ \beta_3^* \end{bmatrix} = \begin{bmatrix} \sum y^{(i)} \\ \sum x_1^{(i)} y^{(i)} \\ \sum x_2^{(i)} y^{(i)} \\ \sum x_1^{(i)} x_2^{(i)} y^{(i)} \end{bmatrix}}$$

ACM/IDS 104 - Problem Set 4 - MATLAB Problems

Before writing your MATLAB code, it is always good practice to get rid of any leftover variables and figures from previous scripts.

```
clc; clear; close all;
```

Problem 3 (10 points) Application of Least Squares to Data Fitting

The dataset `carbig`, a built-in MATLAB dataset, contains various characteristics for $m = 406$ automobiles from the 1970s and 1980s. Let x_1, x_2 and y be the `Weight`, `Horsepower` and `MPG (Miles Per Gallon)` respectively. Let us assume the following theoretical model for the data:

$$y = f(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 \quad (\star)$$

Let $\beta^* = \left(\beta_0^*, \beta_1^*, \beta_2^*, \beta_3^*\right)^T$ denote the best fit to the data, i.e. the vector that minimizes the Euclidean norm of the residual vector $r = (r_1, \cdots, r_m)^T$, where:

$$r_i = y^{(i)} - f\left(x_1^{(i)}, x_2^{(i)}\right)$$

Part (a) (5 points)

Derive the system of normal equations on β^*. Do this in your written-up solutions.

Part (b) (5 points)

Find β^* by solving the normal system numerically and plot the scatter plot of the data $\left\{\left(x_1^{(i)}, x_2^{(i)}, y^{(i)}\right)\right\}, i = 1 \cdots m$, together with the fitted surface $y = f(x_1, x_2)$ given by $(\star)$.

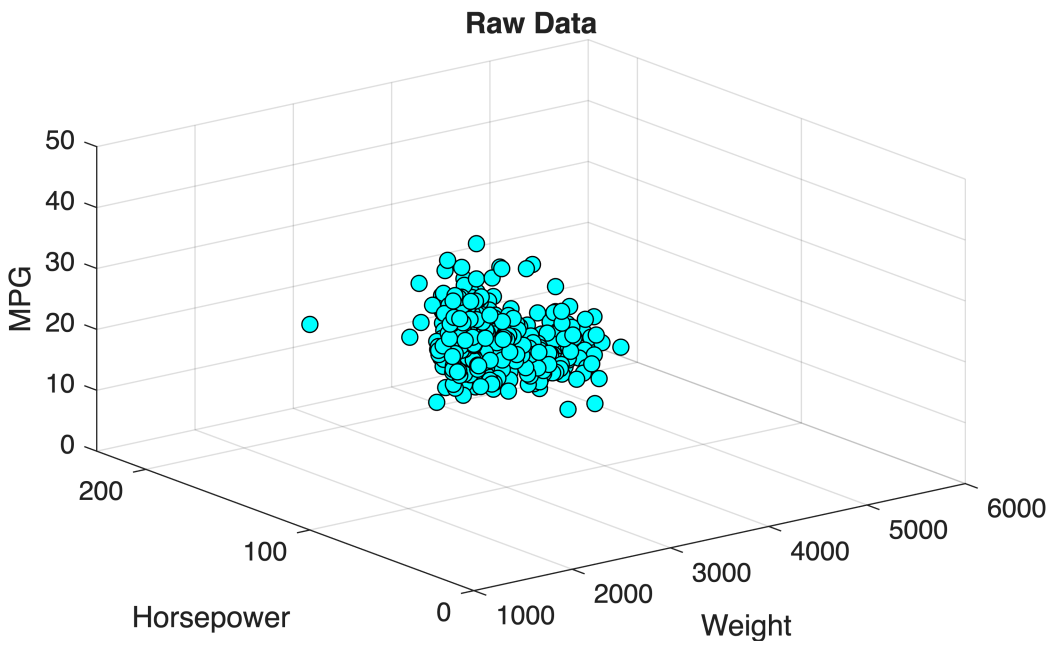
To start, let us load the dataset, define our variables, and perform some necessary cleanup as there exist `NaN` values. We do this for you :)

```
load carbig;
x1=Weight;
x2=Horsepower;
y=MPG;
clearvars -except x1 x2 y;
% Data cleaning
y=y(x1>0);
x2=x2(x1>0);
x1=x1(x1>0);

y=y(x2>0);
x1=x1(x2>0);
x2=x2(x2>0);

x1=x1(y>0);
x2=x2(y>0);
```

```
y=y(y>0);
```

Now, the dataset is ready for you to solve the normal system. Let us vizualize it. Drag the 3D plot around to get a better sense of the data.

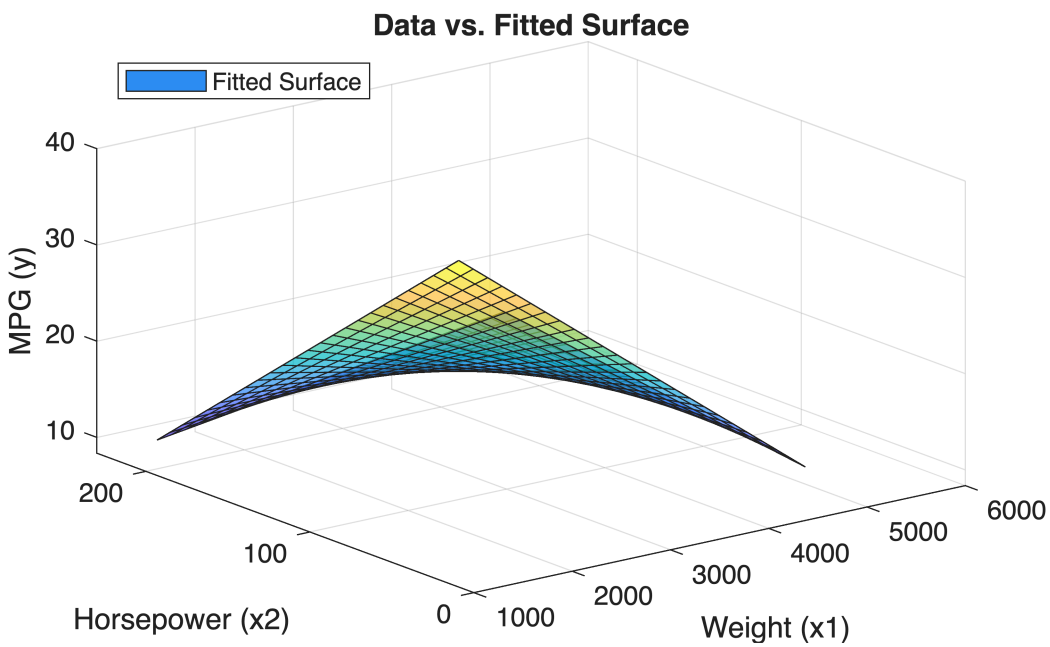
```
figure;
scatter3(x1,x2,y,'MarkerEdgeColor','k','MarkerFaceColor','c');
xlabel('Weight');
ylabel('Horsepower');
zlabel('MPG');
title('Raw Data');
```



Finally, let us find the best surface to fit the data. Remember, in MATLAB, when solving a system, backslash is your best friend. Use `scatter3()` and `fsurf()` when plotting.

```
%{
Build the matrix A
Solve for beta
%}

% num of data points in m
m = length(y);

% A has columns [1, x1, x2, x1*x2]
A = [ones(m, 1), x1, x2, x1 .* x2];

% find least squares solution B* that minimizes norm(A*beta - y).
beta = A \ y;

%{
Plotting
```

```
%}
figure;

% grid of points spanning the range of x1 and x2
x1_fit = linspace(min(x1), max(x1), 25);
x2_fit = linspace(min(x2), max(x2), 25);
[X1_fit, X2_fit] = meshgrid(x1_fit, x2_fit);

% fitted y values on grid
Y_fit = beta(1) + beta(2) * X1_fit + beta(3) * X2_fit + beta(4) *
(X1_fit .* X2_fit);

surf(X1_fit, X2_fit, Y_fit, 'FaceAlpha', 0.7, 'DisplayName', 'Fitted
Surface');

xlabel('Weight (x1)');
ylabel('Horsepower (x2)');
zlabel('MPG (y)');
title('Data vs. Fitted Surface');
legend('Location', 'northwest');
grid on;
hold off;
```



Problem 4 (10 points) Polynomial Interpolation

Interpolating polynomials $p(x)$ are used for approximation of complex functions $g(x)$. Intuitively, the larger the degree of the interpolating polynomial, the more accurate the approximation $g(x) \approx p(x), x \in [a, b]$. Generally, this is not true (as you will see in this problem). High degree interpolating polynomials often behave badly,

especially near the ends of the interval $[a, b]$. In practice, piecewise cubic splines are often used instead of high degree polynomials.

Suppose we want to approximate the function:

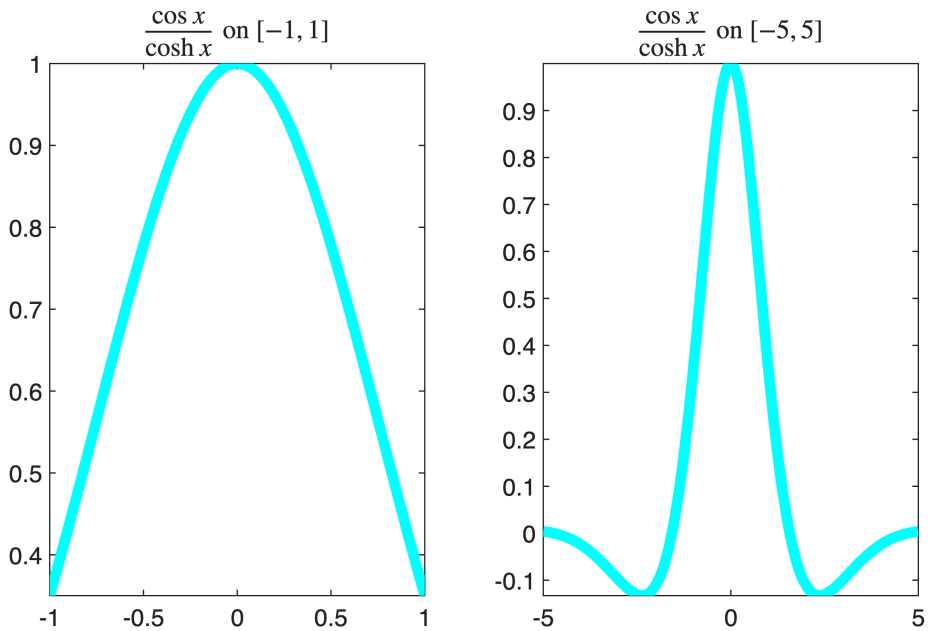
$$g(x) = \frac{\cos x}{\cosh x}, \ x \in [-a, a]$$

using n points equally spaced between $-a$ and a. Find the interpolating polynomials and plot them versus the function $g(x)$ for $a = 1, 5$ and $n = 3, 5, 10, 15$. In your plot, show also the data points used for interpolation. The code should produce a single figure with 8 subplots.

Useful MATLAB functions for this problem:

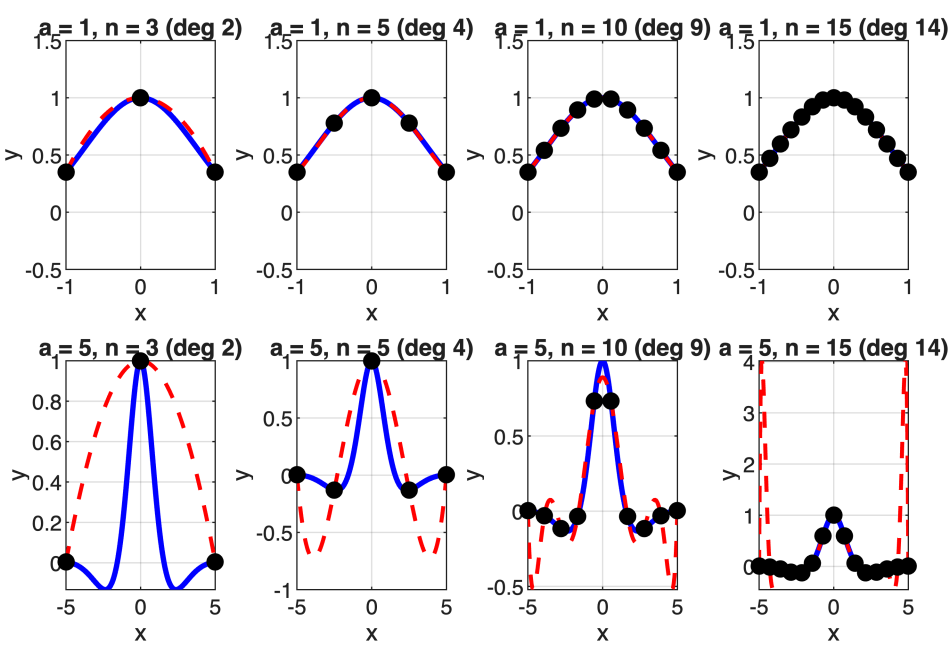
`linspace(), fplot(), polyval(), polyfit()`

```
%{
Let us start by doing some setup
%}
g = @(x) cos(x) ./ cosh(x); % you can define functions like this
a = [1, 5]; % setting interval values
n = [3, 5, 10, 15]; % setting the number of points
sub = 1; % setting the subplot index

%{
Let us see how g(x) looks like on both intervals
%}
figure;
subplot(1, 2, 1);
fplot(g, [-a(1), a(1)], "-c", "lineWidth", 4);
title("$\frac{\cos{x}}{\cosh{x}}$ on $[-1, 1]$","Interpreter","latex");
hold on
subplot(1, 2, 2);
fplot(g, [-a(2), a(2)], "-c", "lineWidth", 4);
title("$\frac{\cos{x}}{\cosh{x}}$ on $[-5, 5]$","Interpreter","latex");
```




```
%{
Complete the nested for-loops
%}
figure;
for ival = a
    for degree = n-1
        n_points = degree + 1;
        subplot(2, 4, sub);

        %{
        Select degree+1 points in the interval
        Evaluate g(x) on these points
        Find the polynomial coefficients
        %}

        x_points = linspace(-ival, ival, n_points);
        y_points = g(x_points);

        p_coeffs = polyfit(x_points, y_points, degree);

        %{
        PLOTTING
        Plot g(x), the sampled points and interpolating polynomial
        Please use different colors and linestyles
        %}

        x_fine = linspace(-ival, ival, 200);

        plot(x_fine, g(x_fine), 'b-', 'LineWidth', 2);
```

```
        hold on;

        y_poly = polyval(p_coeffs, x_fine);
        plot(x_fine, y_poly, 'r--', 'LineWidth', 1.5);

        plot(x_points, y_points, 'ko', 'MarkerFaceColor', 'k');

        title(['a = ' num2str(ival) ', n = ' num2str(n_points) ' (deg '
num2str(degree) ')']);
        xlabel('x');
        ylabel('y');

        if ival == 1
            ylim([-0.5, 1.5]);
        else
            ylim('auto');
        end

        grid on;
        hold off;

        sub = sub + 1;

    end
end
```



Problem 5 (10 points) Stability of the Gram-Schmidt Algorithm

The classical Gram-Schmidt algorithm is numerically unstable. This means that, when implemented on a computer, the round-off errors can cause the output vectors to be significantly non-orthogonal. To explore the issue, perform the following computations for each $n = 10, 20, 30, \cdots, 100$:

1. Create the Hilbert matrix H_n of size n (using `hilb(n)`) and consider the columns $h_1, \cdots, h_n$ as a basis of $\mathbb{R}^n$. The matrix H_n is non-singular, and thus its columns indeed form a basis, but it is very close to singular (i.e. its columns are close to being linearly dependent), and this leads to numerical problems.
2. Implement the basic Gram-Schmidt algorithm to construct an orthogonal basis $\{v_1, \cdots, v_n\}$ of $\mathbb{R}^n$ from $h_1, \cdots, h_n$. Please don't use any advanced built-in function for orthogonalization (such as `orthog()`) just basic matrix operations. At the end of the process normalize your vectors so that the basis is orthonormal.
3. If vectors $v_1, \cdots, v_n$ obtained in (2) are orthonormal, then $V = [v_1, \cdots, v_n]$ must be orthogonal. As a measure of orthogonality, compute the infinite norm $\delta_V(n) = ||I_n - V^T V||_\infty$, which is a matrix norm (use `norm(A, Inf)`). The closer $\delta_V(n)$ is to zero, the closer the columns of V are to being orthogonal.
4. Repeat (2) and (3) to construct an orthonormal basis $\{u_1, \cdots, u_n\}$ using the modified Gram-Schmidt algorithm (which is numerically more stable) described in lecture 9 (page 46), and compute $\delta_U(n)$.
5. To compare the basic and modified Gram-Schmidt algorithms, plot $\delta_V(n)$ and $\delta_U(n)$ versus n.

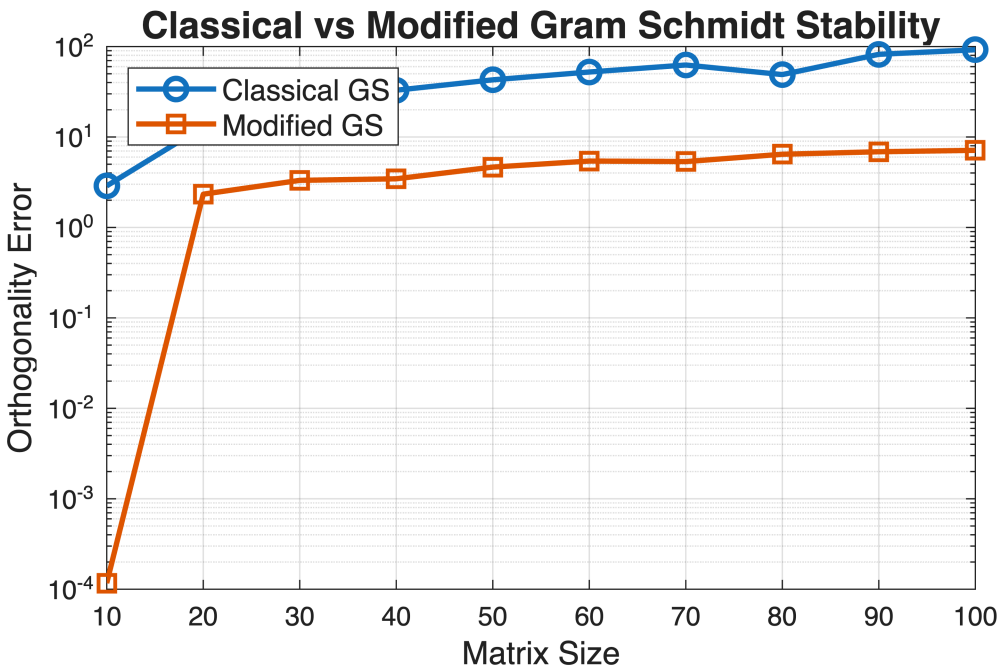
```
%{
Complete the following for-loop following (1)-(4)
%}
n_values = 10:10:100;
delta_V = zeros(size(n_values));
delta_U = zeros(size(n_values));
idx = 1;

for n=10:10:100
    H = hilb(n);
    V = zeros(n, n);

    for j = 1:n
        % jth col of H
        v_j = H(:, j);

        % orthogonalize
        for i = 1:(j-1)
            v_j = v_j - (V(:, i)' * H(:, j)) * V(:, i);
        end

        % normalize
        V(:, j) = v_j / norm(v_j);
    end

    %% orthogonality measure for GS
    I_n = eye(n);
```

```
    delta_V(idx) = norm(I_n - V' * V, Inf);

    % stors orthonormal vectors
    U = zeros(n, n);

    for j = 1:n
        u_j = H(:, j);

        % orthogonalize; project against the updated u_j here instead
        for i = 1:(j-1)
            u_j = u_j - (U(:, i)' * u_j) * U(:, i);
        end

        % normalize
        U(:, j) = u_j / norm(u_j);
    end

    delta_U(idx) = norm(I_n - U' * U, Inf);

    fprintf('n = %3d: δ_V = %.4e, δ_U = %.4e\n', n, delta_V(idx),
delta_U(idx));

    idx = idx + 1;
end
```

```
n =  10: δ_V = 2.8816e+00, δ_U = 1.1675e-04
n =  20: δ_V = 1.3000e+01, δ_U = 2.3324e+00
n =  30: δ_V = 2.2263e+01, δ_U = 3.3172e+00
n =  40: δ_V = 3.2915e+01, δ_U = 3.4510e+00
n =  50: δ_V = 4.2885e+01, δ_U = 4.6494e+00
n =  60: δ_V = 5.2317e+01, δ_U = 5.4218e+00
n =  70: δ_V = 6.2538e+01, δ_U = 5.3499e+00
n =  80: δ_V = 4.9000e+01, δ_U = 6.4467e+00
n =  90: δ_V = 8.2084e+01, δ_U = 6.8704e+00
n = 100: δ_V = 9.2175e+01, δ_U = 7.1246e+00
```

```
%{
Visualize the comparison as specified in (5)
%}
figure;
semilogy(n_values, delta_V, '-o', 'LineWidth', 2, 'MarkerSize', 8,
'DisplayName', 'Classical GS');
hold on;
semilogy(n_values, delta_U, '-s', 'LineWidth', 2, 'MarkerSize', 8,
'DisplayName', 'Modified GS');
hold off;

xlabel('Matrix Size', 'FontSize', 12);
ylabel('Orthogonality Error', 'FontSize', 12);
title('Classical vs Modified Gram Schmidt Stability', 'FontSize', 14);
legend('Location', 'northwest', 'FontSize', 11);
```

```
grid on;
```




```
% Classical GS is numerically unstable
% Modified GS is more stable
```